Isaiah Coleman (he/him)

Backend Engineering Lead | Platform Engineer | Kubernetes & Helm | Hybrid Cloud & On-Prem | Early-stage startups | Automation

Senior Web Application Developer - Duke University Libraries 2024-present

Agile Lead & Platform Engineer *Rails + Hotwire, Python, Go, Postgres*

- Built GitLab CI/CD pipelines and shared application Helm charts, enabling engineering teams to deploy applications and common services to OpenShift.
- Built and maintained reusable CloudFormation templates to provision AWS capacity on demand for research data-processing workloads.
- Modernized a 12+-year-old papyri editing platform for OpenShift, refactoring the application from a monolith run on VM and recomposing it as discrete application services managed by Helm charts for rootless Kubernetes deployments.
- Integrated Prometheus metrics, centralized application logs in Splunk, used Airbrake/Erribit to track exceptions and built Grafana dashboards and alerts to track CPU and memory utilization, identify recurring resource spikes, triage exceptions and recurring errors and resolve performance bottlenecks through targeted code fixes.
- Led incident response during outages and surges in high-volume bot traffic with +7000% traffic volume; deployed Anubis to filter bot requests and used Matomo and Grafana to monitor traffic patterns and system load.
- Used Claude and locally hosted models via ollama in agentic development workflows to accelerate implementation, critically review AI-generated code, and deliver production-ready software through rigorous testing and validation.

Web Application Developer - NCSU University Libraries 2023 - *2024*

PHP, Angular, Postgres

- Integrated Mazevo with the campus-wide room-reservation application, retaining the existing user interface while shifting selected scheduling rules and event storage to the platform.

Lead Backend Developer - Crosscomm 2022-2023

Node.js, Ruby on Rails, React, Postgres

- Designed and built APIs and context-aware recommendation services for the American Cancer Society mobile app using Next.js, and integrated OAuth-based authentication and identity management with Amazon Cognito.
- Built a self-service AWS job orchestration platform for higher-education research labs, automating HPC job creation, Slurm queue provisioning and teardown, and output delivery to Amazon S3.
- Established coding standards, defined project roadmaps and deliverables, and translated client business needs into actionable engineering plans.
- Early adopter of AI-assisted development workflows, using ollama with Continue and GitHub Copilot to accelerate implementation, generate boilerplate, and review code; validated changes through testing before production.

Senior Integration Engineer - LearnPlatform (Instructure) 2019 - 2022

Rails, Elixir, React, postgres

- Joined as early in-house engineer and lead, transitioning product development from external contractors to an internal team; established coding standards and release practices, and mentored junior engineers through office hours and developer show-and-tells.
- Built distributed Elixir integration systems processing 30 billion events annually from Chrome extensions deployed across school districts via Google Admin policies and MDM.
- Designed event-driven pipelines integrating Chrome extension data and student information system (SIS) data using AWS EventBridge, Lambda, SQS, and SNS.
- Implemented scheduled, secure student information system (SIS) data transfers to Amazon S3 using AWS Transfer Family, with S3 events routed through SNS and SQS to trigger processing by an Elixir backend.

Software Engineer - SAS Institute 2013 - 2019

Java, React, Mysql

- Designed and implemented SAML, OAuth 2.0, and enterprise SSO integrations, including automated identity provisioning, account lifecycle management, and role-based permission mapping, supporting more than 4 million users
- Architected, developed, and released Objective-C iOS apps - SAS Match Stretch and SAS Data Notebooks

Contact:
isaiahccoleman@gmail.com
iccole.dev
github.com/iccole
linkedin.com/in/isaiah-coleman

Education:
North Carolina State University
B.S. in Computer Science
Minor - Cognitive Science

Skills:

Programming Languages: Elixir, Ruby, Typescript, Objective-C, Clojure, Javascript, Go, Python

Cloud & Infrastructure: AWS, Azure, GCP, Red Hat OpenShift (OKD) Hybrid Cloud & On-Prem

Infrastructure as Code: Terraform, Helm, Ansible, CloudFormation

Observability: Grafana, Prometheus, Splunk, Airbrake, Zabbix, Rollbar, Matomo

System Administration: Debian, Red Hat Enterprise Linux, AlmaLinux

Core Areas: Software Design & Implementation, Release Management, Git Workflow Design & Software Governance, API Design, Algorithms & Data Structures, Performance Optimization & Reliability, System Integrations & Interoperability, Agile/Kanban

Databases & Messaging: Postgres, MySQL, Redis, AWS Redshift, AWS Kinesis, AWS SNS, AWS SQS, AWS RDS